

Explainable Survival Prediction from Gene Expression and Clinical Data

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ABSTRACT

Doctors usually estimate how long a cancer patient will live from the size and spread of the tumour. Two patients placed in the same group often live for very different lengths of time. Measurements of gene activity can explain part of that difference. Models built on such measurements are usually hard to read, so a model was built here that stays readable at every step. Clinical records and gene activity were combined for 1,620 patients from four public cancer datasets. The datasets cover brain, liver, pancreatic and skin cancer. Highly similar genes were removed first, which reduced 500 candidate genes to 59. These 59 genes were compressed into 50 summary variables using principal component analysis, a standard method for reducing the number of variables. A Cox proportional hazards model was then fitted with an elastic-net penalty, which shrinks unhelpful coefficients towards zero. On a held-out test set of 324 patients the model reached a concordance index of 0.736. The concordance index measures how often two patients are ranked in the correct survival order. Raw risk scores were converted into survival probabilities using isotonic regression. Brier scores, which measure probability error, ranged from 0.132 to 0.178 across the 12, 24, 36 and 60-month horizons. Uncertainty for each patient was estimated from 5,000 simulated survival times. The compression step was reversed by matrix multiplication, which returned an exact risk weight for every one of the 59 genes. No approximate explanation method was needed.

KEYWORDS

Survival analysis; gene expression; Cox proportional hazards; elastic net; probability calibration; model explainability

1 Introduction

Cancer treatment decisions depend on an estimate of how long a patient is likely to live. That estimate is usually built from the tumour, node and metastasis (TNM) staging system [1]. The system groups patients by tumour size and by how far the disease has spread. Staging describes anatomy. It does not describe the molecular activity inside the tumour. Two patients placed in the same stage therefore often survive for very different lengths of time. Part of that gap can be traced to differences in gene activity. Ribonucleic acid sequencing (RNA-Seq) measures how active each gene is in a tissue sample [2]. Some genes measured this way are already linked to poor outcomes. High activity of the gene CXCL5 is associated with shorter survival in liver cancer [3]. High activity of the gene CA9 marks tumour regions starved of oxygen, and is also linked to worse outcomes [4]. The Cancer Genome Atlas (TCGA) has released matched clinical records and gene activity measurements for dozens of cancer types [5]. This raises a direct question. Can staging information and gene activity be combined into one model that predicts survival better than staging alone, and that a doctor can still inspect?

The standard tool for this task is the Cox proportional hazards model [6]. The model works with a hazard, which is the instantaneous risk of dying at a given moment for a patient who is still alive. The Cox model does not assume any particular shape for how that baseline risk changes over time. It only assumes that each patient's hazard is the baseline hazard multiplied by a fixed factor. That factor comes from a weighted sum of the patient's measurements. The model also handles right censoring, which occurs when a patient is still alive when the study ends. Such a patient contributes the information that they survived at least that long. This flexibility comes with a limit. The number of measurements must stay well below the number of patients. When the number of measurements approaches the number of patients, the fitted weights are no longer uniquely determined [7]. Gene activity data breaks this limit immediately, because a single experiment measures thousands of genes for a few hundred patients.

Three families of methods have been used to work around this limit. The first family adds a penalty term to the Cox model that pushes weights towards zero. The lasso penalty sets some weights exactly to zero and so selects a subset of genes [7]. The elastic-net penalty adds a second term to that behaviour [8, 9]. The second term shrinks correlated genes together, instead of picking one gene from each correlated group at random [10]. Standard textbook treatments

of both penalties are available [11]. The second family reduces the number of variables before fitting. Principal component analysis (PCA) replaces many correlated measurements with a few summary variables that capture most of the variation [12, 13]. Cross-validated penalised Cox models applied to gene activity data follow this pattern [14]. The third family corrects the probabilities a model reports. Isotonic regression maps a raw score onto a probability [15]. It preserves the ordering of patients and assumes nothing about the shape of that mapping [16]. Separately, neural network models such as DeepSurv [17], Cox-nnet [18] and Cox-PASNet [19] replace the weighted sum with a flexible function. These models can capture effects that a weighted sum cannot. They also need more patients than a single cancer cohort usually provides, and their internal weights no longer map onto named genes.

Four specific problems remain unsolved when these methods are used together. First, groups of genes that act in the same biological pathway are often left in the model together. Their measurements are almost identical, which makes the fitted weights unstable across repeated fits [10]. Second, the Cox model reports a relative risk score. A doctor cannot read a one-year or five-year survival probability off that score without a separate conversion step, and that step is frequently skipped. Third, a single predicted number says nothing about the range of outcomes that are plausible for one patient. That range matters for surgical and palliative care planning. Fourth, PCA solves the variable-count problem but creates a readability problem. The fitted weights attach to summary variables, not to named genes. Post-hoc explanation methods such as SHapley Additive exPlanations (SHAP) [20] and Local Interpretable Model-agnostic Explanations (LIME) [21] estimate gene-level importance by sampling. They approximate the answer rather than recovering it exactly.

This work addresses all four problems in a single model. Five contributions are reported. A correlation filter removes redundant genes using the training data only, reducing 500 candidate genes to 59. A Cox model with an elastic-net penalty is fitted using a smoothed objective, which keeps the optimiser stable where the penalty is otherwise not differentiable. Isotonic regression calibrators, fitted on a separate split, convert risk scores into survival probabilities at four time horizons. A simulation draws 5,000 possible survival times per patient, giving pessimistic, median and optimistic estimates. Finally, the PCA step is reversed by one matrix multiplication. This returns an exact risk weight for each of the 59 genes. It also returns an exact additive breakdown for each patient. Figure 1 shows how data moves through these stages.

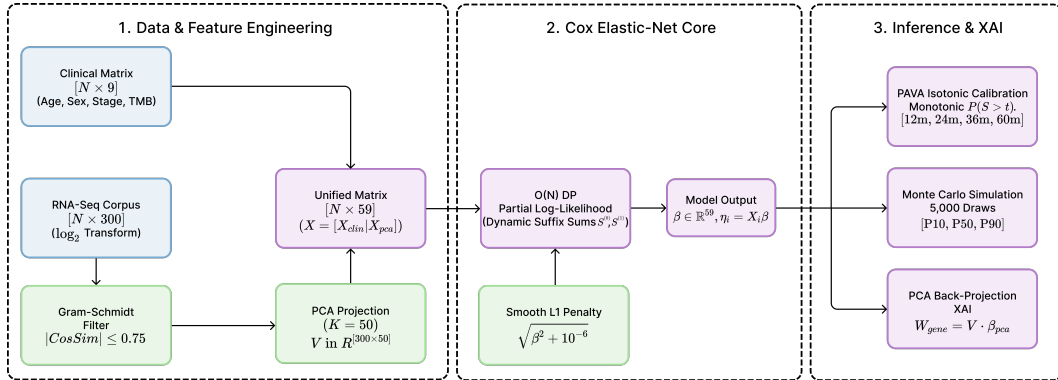


Figure 1: Overall data flow. Raw clinical records and gene activity measurements enter on the left. They pass through the correlation filter, the compression step, the penalised Cox model, the probability calibrator and the simulation stage. Gene-level explanations leave on the right.

2 Related Work

2.1 Statistical models

Early work on gene activity and survival extended the Cox model with penalty terms. Ridge, lasso and elastic-net penalties were all applied to the Cox partial likelihood [7–10]. These methods keep the weighted-sum structure of the Cox model, so each fitted weight still refers to one named measurement. Cross-validated penalised Cox regression on microarray data established the practical recipe still in use [14]. That recipe fits the penalty strength by cross-validation. Performance is then reported on data the model never saw. Random survival forests took a different route [22]. They drop the assumption that each patient’s hazard is a fixed multiple of the baseline hazard. Error is estimated from the samples left out of each tree.

2.2 Neural network models

DeepSurv replaced the weighted sum in the Cox model with a neural network [17]. Cox-nnet applied the same idea to gene activity data from ten TCGA cohorts [18]. Cox-PASNet added biological pathway structure to the network, and was tested on TCGA brain and ovarian cancer data [19]. DeepProg combined several deep and classical models into an ensemble across many TCGA cancer types [23]. It is notable for testing on independent cohorts rather than by cross-validation alone. Dynamic-DeepHit went further [24]. It uses repeated measurements over time to update its predictions as new data arrives. Across this group, the concordance index is almost always the headline metric. Calibrated probabilities, per-patient uncertainty and external validation are reported far less consistently.

2.3 Feature representation

Most models in this area use gene activity data together with clinical variables. Some add measurements of microRNA, deoxyribonucleic acid methylation or tissue images. Autoencoder models compress several such data types into a small set of learned variables. Chaudhary and colleagues applied this approach to 360 TCGA liver cancer patients [25]. They reported a concordance index of 0.68, and tested on independent liver cancer cohorts. Learned variables of this kind can capture patterns a weighted sum cannot. They are also difficult to trace back to individual genes, which is the readability gap this work targets.

2.4 Comparison of existing models

Table 1 summarises the studies discussed above.

Table 1: Summary of related survival prediction studies.

Study	Cohort	Input data	Model	Validation and metric
van Houwelingen et al. [14]	Breast cancer, microarray	Gene expression	Penalised Cox	Cross-validation
Ishwaran et al. [22]	General survival data	Clinical and mixed	Random survival forest	Out-of-bag error
Katzman et al. [17]	Clinical cohorts	Clinical data	DeepSurv	Concordance index
Ching et al. [18]	Ten TCGA cohorts	Gene expression	Cox-nnet	Concordance index
Hao et al. [19]	TCGA brain and ovarian	Expression, clinical, pathway	Cox-PASNet	Concordance index
Chaudhary et al. [25]	TCGA liver, $n = 360$	Expression, microRNA, methylation	Autoencoder	Concordance index 0.68; independent cohorts
Poirion et al. [23]	32 TCGA cancers	Multiple data types	DeepProg	Concordance index; independent cohorts
Lee et al. [24]	Cystic fibrosis registry, $n = 5,883$	Repeated clinical measurements	Dynamic-DeepHit	Time-dependent discrimination
This work	Four TCGA cancers, $n = 1,620$	Expression and clinical	Cox elastic net	Held-out test split; concordance index, area under the curve, Brier score

2.5 Limitations of existing approaches

Several limitations recur across this literature. Sample sizes are small relative to the number of measurements, which raises the risk of unstable weights. TCGA records one tissue sample per patient at diagnosis, so models that need repeated measurements over time cannot be applied directly [24]. Forcing a time-varying structure onto single-timepoint data requires carrying values forward artificially, which introduces immortal time bias [26]. Learned variables from autoencoders are hard to map back to named genes [25]. Pooling several cancer types into one model averages gene effects that may differ between those types. Most importantly, discrimination is reported far more often than calibration or per-patient uncertainty, and external validation remains inconsistent.

2.6 Research gap

Penalised Cox regression [7, 10], variable compression [14], probability calibration [27], uncertainty estimation [24] and gene-level explanation [19] have each been studied separately. No published pipeline combines all five at once. Such a pipeline must also keep the training, calibration and test data strictly separated. It must return an exact gene-level explanation rather than an approximate one. That combination is the contribution of this work. The current model uses one measurement per patient, taken at diagnosis. A version using repeated measurements is left for future work.

3 Method

3.1 Cohort assembly

Patients were drawn from four TCGA cohorts. The first was glioblastoma multiforme (TCGA study code GBM), a brain cancer [28]. The second was liver hepatocellular carcinoma (TCGA study code LIHC) [29]. The third was pancreatic adenocarcinoma (TCGA study code PAAD) [30]. The fourth was skin cutaneous melanoma (TCGA study code SKCM) [31]. The codes GBM, LIHC, PAAD and SKCM are dataset identifiers used by TCGA, not standard clinical abbreviations. Clinical records and RNA-Seq gene activity values were downloaded in the file format published by cBioPortal [32, 33]. The four cohorts were merged into one table of $N = 1,620$ patients.

Clinical fields that exist in only one cohort were removed. The melanoma skin type field is one example. Such a field is missing for every patient in three of the four cohorts. Filling those gaps with a single value would create an artificial cluster of identical entries.

The cohort was split into three parts. Training used 972 patients, calibration used 324 and testing used 324. The split was stratified on whether a death was observed, rather than on survival time. Stratifying on survival time under right censoring shifts the distribution between splits. Stratifying on the observed event instead held the death rate nearly constant: 58.4% in training, 58.6% in calibration and 58.3% in testing.

3.2 Removing redundant genes

The 500 candidate gene activity columns were screened before any scaling or compression. The absolute Pearson correlation was computed between every pair of those 500 columns, using the training split only. Columns were then examined in a fixed order. A column was discarded if its absolute correlation with any earlier column exceeded 0.75. This rule keeps the first member of each correlated group and discards the rest.

On this cohort, 36,205 gene pairs exceeded the threshold. The filter discarded 441 of the 500 genes and kept 59. Discarded genes were not arbitrary. The strongest pairs were genes known to act together. The two acyl-coenzyme A synthetase genes ACSM2A and ACSM2B correlated at $r = 0.996$. The three fibrinogen chain genes FGA, FGB and FGG correlated above $r = 0.98$ with each other. This pattern indicates that the filter removed genuine biological redundancy rather than random noise. The screen costs one correlation matrix. It avoids the repeated matrix inversions that a full variance inflation factor audit would require. Figure 2 shows the correlation structure of a representative subset of genes before and after this step.

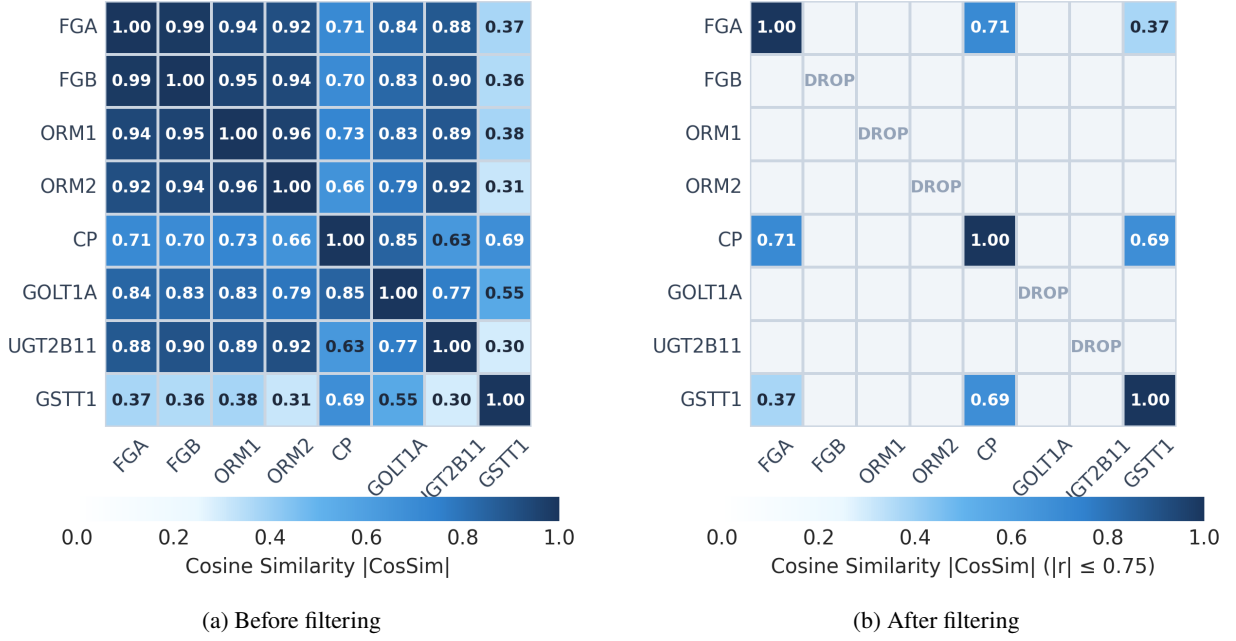


Figure 2: Pairwise correlation between a representative subset of genes, measured on the training split. Dark cells on the left mark groups of genes that move together, such as the fibrinogen chain genes. On the right, the discarded genes are marked and only weakly correlated genes remain.

3.3 Scaling and compression

RNA-Seq values are counts. Their distribution has a long right tail, because a few genes are far more active than the rest [34]. A base-two logarithm, $\log_2(x + 1)$, was applied to the 59 surviving columns to shorten that tail. The added one prevents an undefined logarithm where activity is zero.

The transformed values were then scaled to have a mean of zero and a standard deviation of one. This produced the matrix $Z_{\text{gene}} \in \mathbb{R}^{N \times 59}$. The scaling was computed inside each cross-validation training fold only. Computing it on all data first would leak information from the validation fold into training.

Scaling has to happen before compression. PCA finds the directions of largest variation [12, 13]. Without scaling, a gene whose numbers simply span a wider range would dominate the first direction, whatever its biological role. The loading matrix $V \in \mathbb{R}^{59 \times 50}$, which satisfies $V^T V = I_{50}$, was computed by singular value decomposition. The scaled genes were projected onto $K = 50$ summary variables, giving $X_{\text{pca}} = Z_{\text{gene}} V \in \mathbb{R}^{N \times 50}$. Figure 4 shows how the surviving genes map onto the leading summary variables.

3.4 Model input and objective

Clinical fields were processed on a separate branch. Sex, race, ethnicity, cancer type and age group were one-hot encoded, which turns each category into its own binary column. Age was filled with the median where missing, then scaled. This produced a clinical block $X_{\text{clin}} \in \mathbb{R}^{N \times 20}$. The clinical block and the 50 summary variables were joined and rescaled, giving the model input $X = [X_{\text{clin}} \mid X_{\text{pca}}] \in \mathbb{R}^{N \times 70}$.

The Cox model assigns each patient i a risk score $\eta_i = X_i \beta$, with hazard $h(t \mid X_i) = h_0(t) \exp(\eta_i)$ [6]. The weights are $\beta = [\beta_{\text{clin}} \mid \beta_{\text{pca}}] \in \mathbb{R}^{70}$. Fitting requires the Breslow partial log-likelihood [35]. Evaluated directly, this needs a nested loop over patients at every observed death, which costs $O(N^2)$ operations. Death times were instead sorted in descending order, and running totals of the exponentiated risk scores were accumulated in a single pass. This reduces the cost to $O(N)$.

An elastic-net penalty was added to the negative partial log-likelihood [8, 9]. The penalty has an absolute-value term and a squared term. The absolute-value term drives weights of uninformative variables to zero. The squared term shrinks correlated variables together.

3.5 Optimisation

The objective was minimised with the limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with bound constraints (L-BFGS-B) [36]. This method needs a gradient everywhere. The absolute-value term has no gradient at zero, and produced undefined values during development. The absolute value was therefore replaced by the smooth approximation $\sqrt{\beta^2 + \epsilon}$ with $\epsilon = 10^{-6}$. That value is small enough to keep the sparsifying behaviour near zero. It is also far enough above double-precision rounding error to keep the gradient stable.

The risk score η_i was clipped to the range $[-40, 40]$ before exponentiation. The upper bound gives $\exp(40) \approx 2.35 \times 10^{17}$, which already represents an extreme relative risk. It stays well below the double-precision overflow point at $\exp(709)$.

Two hyperparameters were tuned by grid search under five-fold nested cross-validation. The penalty strength α was searched over $\{0.1, 0.3, 0.8, 1.5, 3.0\}$. The balance between the two penalty terms, $l1_ratio$, was searched over $\{0.0, 0.1, 0.3, 0.5, 0.7\}$. The search selected $\alpha = 3.0$ and $l1_ratio = 0.7$. The final fit converged after 51 iterations. Figure 3 shows how the fitted weights and loadings feed the later stages. The implementation used scikit-learn for preprocessing and cross-validation [37].

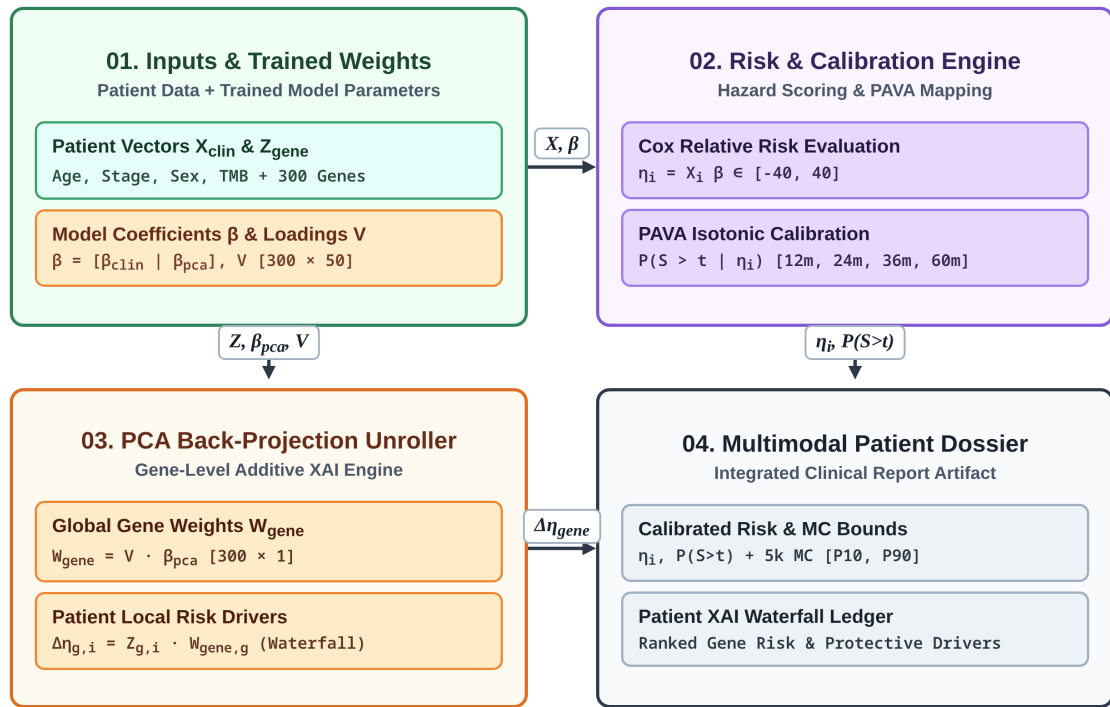


Figure 3: Flow through the prediction and explanation stages. Fitted weights and compression loadings feed the risk score, the probability calibrator and the reversal step that produces gene-level and patient-level explanations.

3.6 Converting risk scores into probabilities

The risk score η_i is only meaningful relative to other patients. It does not answer the question a patient asks, which is the chance of surviving past a given date. The 324-patient calibration split was held out before any filtering or scaling took place. Isotonic regression was fitted on that split [15, 16, 27]. Isotonic regression fits a step function that never decreases, so it cannot reverse the ordering the Cox model already produced. Separate calibrators were fitted at 12, 24, 36 and 60 months. Each maps a risk score to a survival probability, using the observed survival fractions from the Kaplan-Meier estimator as the target [38]. Figure 5 plots the four fitted mappings.

3.7 Estimating uncertainty by simulation

A single predicted number gives no sense of the spread of possible outcomes. Individual survival times were therefore simulated against the Breslow cumulative baseline hazard $\Lambda_0(t)$, estimated from the training split [35]. For each patient, 5,000 independent values $U^{(k)} \sim \text{Uniform}(0, 1)$ were drawn. Each draw was converted to a target hazard, $\Lambda_{\text{target}} = -\ln(U^{(k)})/\exp(\eta_i)$. The matching month was located by binary search over the baseline hazard step function. That lookup costs $O(\log E)$ operations, where $E = 461$ is the number of distinct observed death times.

The 5,000 draws per patient were summarised as a pessimistic bound ($P10$), a median ($P50$) and an optimistic bound ($P90$). The simulated survival curve was also integrated up to 60 months to give the restricted mean survival time [39]. Integration was capped at the longest observed follow-up in the cohort, 218.4 months, so the simulation never extrapolates beyond the data. Figure 6 shows example bands.

3.8 Recovering gene-level weights exactly

PCA reduced 59 genes to 50 summary variables, so the fitted weights attach to those summary variables rather than to genes. Both the compression and the Cox risk score are linear operations. The compression can therefore be reversed exactly.

The sub-vector of weights attached to the summary variables is $\beta_{\text{pca}} \in \mathbb{R}^{50}$. Multiplying the loading matrix by this sub-vector gives a risk weight for every surviving gene:

$$W_{\text{gene}} = V \cdot \beta_{\text{pca}} \in \mathbb{R}^{59}. \quad (1)$$

For patient i and gene g , the contribution to the risk score follows directly:

$$\Delta\eta_{g,i} = Z_{g,i} \cdot W_{\text{gene},g}. \quad (2)$$

Here $Z_{g,i}$ is the scaled activity value that entered the compression step. Because this is exact linear algebra, the contributions of all 70 model inputs add up to the patient’s risk score with no residual. Sorting them by size produces a breakdown that can be checked term by term, shown in Figure 7. Sampling-based methods such as SHAP [20] and LIME [21] estimate the same quantity, but only approximately.

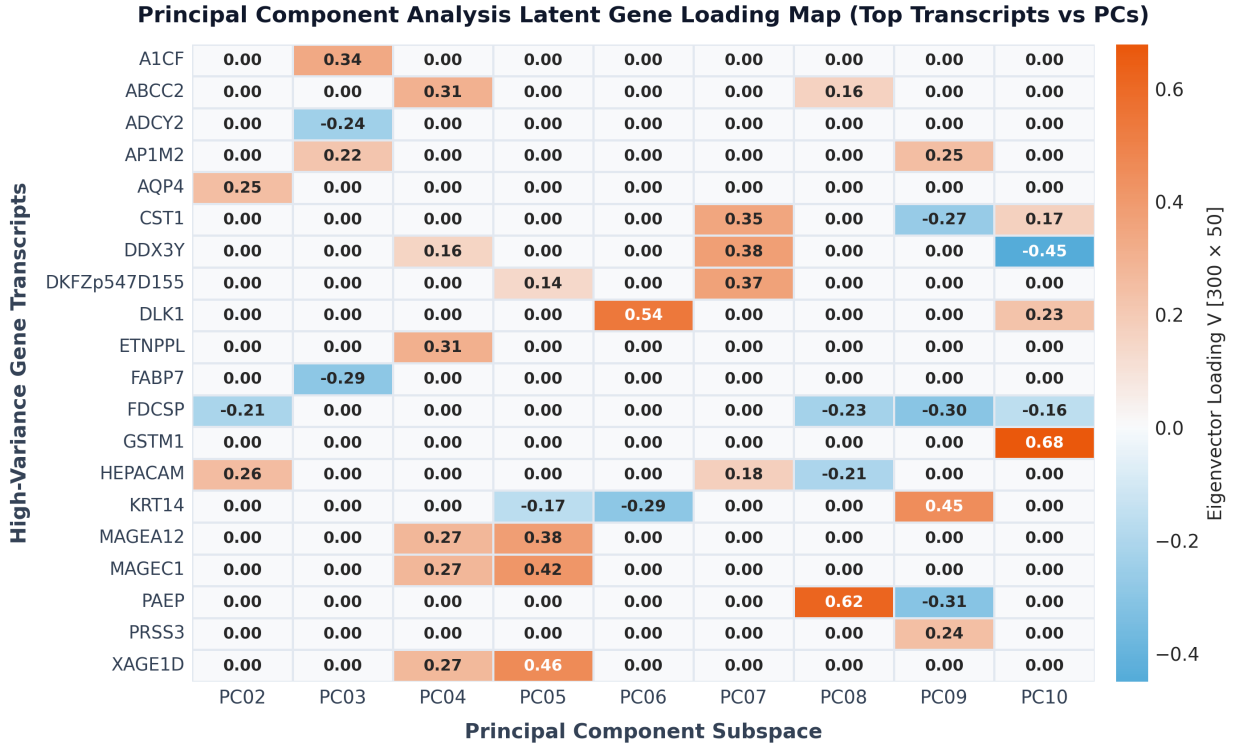


Figure 4: Loadings of a representative subset of the 59 surviving genes on the first ten summary variables. Cell values give the size of each gene’s contribution to the corresponding summary variable.

4 Results

4.1 Feature reduction

The model input shrank in stages. The pipeline started with 506 candidate columns, made up of 500 gene activity columns and 6 clinical fields. The correlation filter left 65 columns, of which 59 were genes and 6 were clinical.

One-hot encoding then expanded the 6 clinical fields into 20 columns, while PCA compressed the 59 genes into 50 summary variables. The final model input had 70 columns. Table 2 lists these stages.

The correlation filter removed 441 of 500 genes, which is 88.2% of the panel. This indicates that most of an unfiltered 500-gene panel carries repeated information rather than independent signal. It supports filtering before compressing, instead of relying on PCA alone to absorb the redundancy.

Table 2: Number of columns at each stage of the pipeline.

Stage	Columns
Candidate gene activity columns	500
Clinical fields, before encoding	6
Gene columns after correlation filter	59
Clinical fields after correlation filter	6
Summary variables after compression	50
Clinical columns after one-hot encoding	20
Final model input X	70

4.2 Model performance

Five-fold nested cross-validation on the training split selected $\alpha = 3.0$ and $l1_ratio = 0.7$. The mean cross-validated concordance index was 0.722, with a standard deviation of 0.021 across folds. The concordance index measures how often the model ranks two patients in the correct survival order [40]. A value of 0.5 matches random ordering, and 1.0 matches perfect ordering. Censoring-adjusted variants of this statistic have also been proposed [41].

The final model reached a concordance index of 0.749 on the training split and 0.746 on the calibration split. On the held-out test split of 324 patients it reached 0.736. Table 3 lists these values. The gap between training and test performance was 0.013. A gap this small indicates limited overfitting, which is consistent with the aggressive filtering and shrinkage applied earlier.

Table 3: Concordance index by data split.

Split	Concordance index
Training ($n = 972$)	0.749
Calibration ($n = 324$)	0.746
Test ($n = 324$)	0.736
Five-fold nested cross-validation (mean $\pm$ s.d.)	0.722 ± 0.021

The fitted weights show which inputs mattered most. Cancer type carried the largest clinical effect. Membership of the brain cancer cohort gave the largest risk-increasing weight ($\beta = 0.772$). Membership of the melanoma cohort gave the largest protective weight ($\beta = -0.322$). This ordering matches the observed survival in these cohorts, reported in Table 5. Patient age gave the third-largest clinical weight ($\beta = 0.293$), increasing risk. The first summary variable gave the largest weight among the compressed gene variables ($\beta = 0.304$).

The two blocks contributed comparable amounts. Summed across all 50 summary variables, the absolute weights of the gene block totalled 2.814. The 20 clinical columns totalled 2.086. Both blocks were scaled the same way, so these totals are directly comparable. By this measure the compressed gene signal carried slightly more weight than the full clinical panel. Figure 7(a) ranks the ten largest weights across both blocks.

4.3 Calibration at fixed time horizons

The calibrators were evaluated on the 324-patient test split. At each horizon the evaluation was restricted to patients whose outcome at that horizon is known, after accounting for right censoring. That gave 289, 247, 228 and 218 patients at 12, 24, 36 and 60 months. Table 4 reports the area under the curve and the Brier score at each horizon.

The area under the curve measures discrimination at a fixed time point [42]. It rose from 0.759 at 12 months to a peak of 0.860 at 24 months, then settled near 0.847 at 36 and 60 months. The Brier score measures the squared difference between the predicted probability and the observed outcome, so lower values are better [43]. It fell from 0.178 at 12 months to 0.132 at 60 months.

The 60-month horizon had both the lowest error and the highest observed death rate, at 78.4%. The death rate rises with horizon length for a mechanical reason. As the window extends, fewer patients remain who were censored before the horizon. More of the uncertainty at 12 months therefore comes from patients last seen alive and event-free. Figure 5 plots the fitted mapping at each horizon. Each curve rises without ever falling, which confirms that the calibrator preserved the ordering the Cox model produced.

Table 4: Calibration results on the held-out test split.

Horizon (months)	Patients with known outcome	Death rate	Area under curve	Brier score
12	289	0.284	0.759	0.178
24	247	0.543	0.860	0.152
36	228	0.662	0.847	0.150
60	218	0.784	0.847	0.132

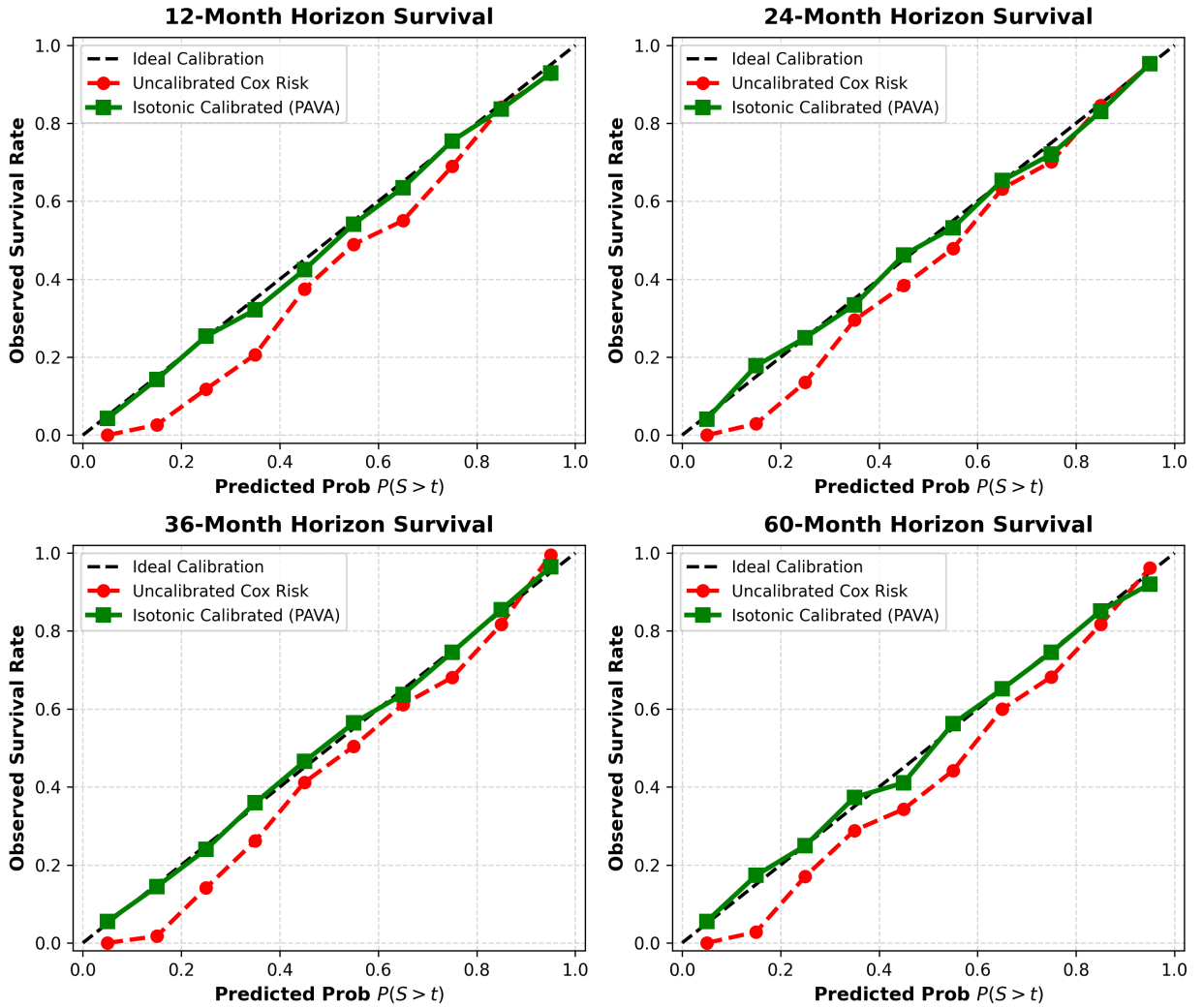


Figure 5: Fitted calibration curves at (a) 12 months, (b) 24 months, (c) 36 months and (d) 60 months. Each panel compares the uncalibrated Cox risk score with the calibrated survival probability on the test split. The diagonal marks perfect calibration.

4.4 Simulated uncertainty

The simulation produced individual P_{10} , P_{50} and P_{90} survival bounds for all 324 test patients. The mean width of the interval between P_{10} and P_{90} was 107.8 months. The median width was 103.0 months. Intervals this wide reflect the difficulty of projecting a single tissue sample forward across a follow-up window reaching 218.4 months.

Interval width varied with risk. One high-risk test patient with $\eta_i = +0.965$ received bounds of $P10 = 3.3$, $P50 = 13.6$ and $P90 = 38.1$ months, with a 60-month restricted mean survival time of 16.9 months. One low-risk test patient with $\eta_i = -1.578$ received much wider bounds of $P10 = 20.8$ and $P50 = 152.2$ months. That patient's $P90$ reached the 218.4-month cap set by the longest observed follow-up, and the restricted mean survival time was 52.3 months.

This asymmetry follows from the exponential form of the hazard. A large positive risk score compresses the simulated death times towards the present, whatever value $U^{(k)}$ takes. A large negative score spreads the same distribution across the whole follow-up range. Figure 6 plots both cases. In practice the $P10$ bound supplies a worst-case timeline for planning aggressive treatment. The $P90$ bound supplies a best-case ceiling that is limited by the cohort's own follow-up rather than by unbounded extrapolation.

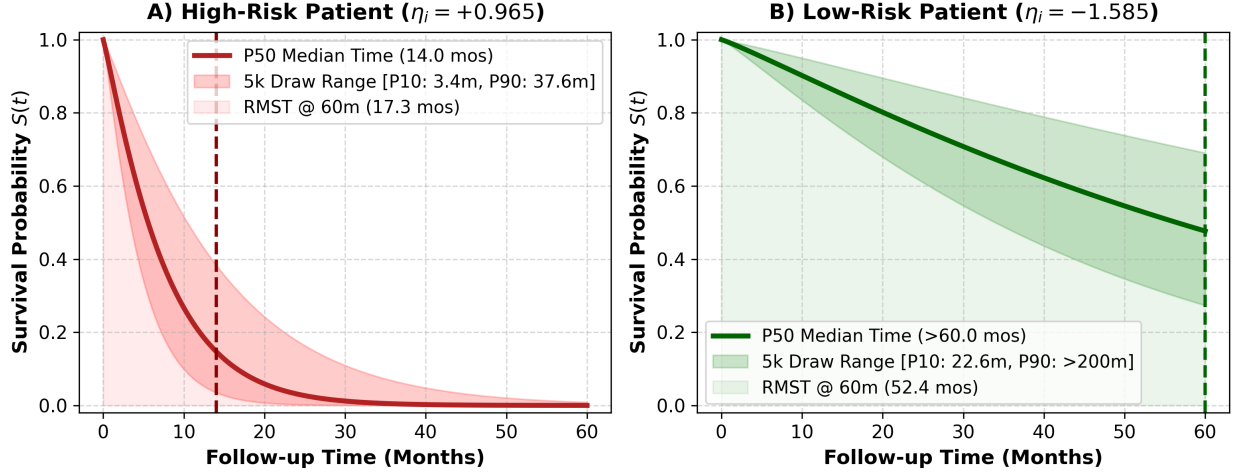


Figure 6: Simulated survival probability over a 60-month window for one high-risk and one low-risk test patient.

4.5 Gene-level and patient-level explanations

Reversing the compression resolved the first summary variable into its constituent genes. The ten genes loading most strongly on that variable were CD24, CXCL14, GPRC5A, PPP1R1B, SYT13, SERPINA3, CA9, ADAM6, FGFR2 and CXCL5. Two of these have documented links to low-oxygen tumour regions and to spread beyond the original site: CA9 [4] and CXCL5 [3].

At the level of one patient, the breakdown splits the risk score across all 70 model inputs with no residual. Figure 7(b) shows a representative test patient, identifier TCGA-DD-AACJ, who was still alive at last contact after 69.0 months of follow-up. For this patient the largest single term was protective: not belonging to the brain cancer cohort, contributing -0.616 . The two largest risk-increasing terms were patient age, contributing $+0.328$, and one summary variable, contributing $+0.321$. All 70 terms summed to the patient's risk score of -0.028 . Because this decomposition is exact matrix algebra, the numbers can be audited individually.

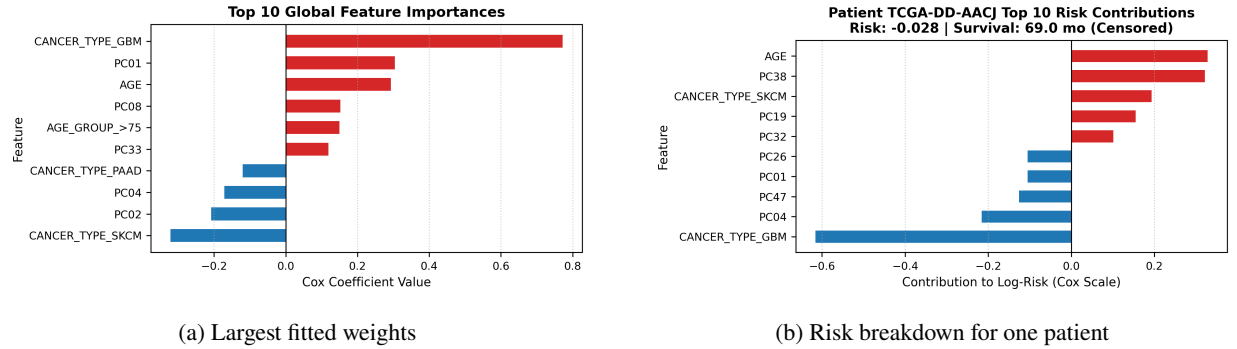


Figure 7: Explanations produced by reversing the compression step. (a) Inputs ranked by fitted weight across the whole cohort. (b) Additive breakdown of one patient's risk score.

4.6 Independent statistical screen

The raw clinical and gene activity columns were also tested for association with mortality, separately from the model. Two tests were used. A two-sided Mann-Whitney U test compared each variable against whether a death was observed [44]. A Spearman rank correlation compared each variable against survival time in months [45]. Both were corrected for multiple testing across all 1,049 evaluated columns of the 1,620-patient cohort, using the Benjamini-Hochberg procedure [46]. Both tests examine one variable at a time, so they show association rather than an effect independent of the other variables.

Cancer type was strongly associated with mortality ($p = 1.09 \times 10^{-53}$, corrected $q = 1.36 \times 10^{-51}$, rank-biserial effect size 0.392). This matches the median survival across the pooled cohort, which ranged from 12.16 months for brain cancer to 35.78 months for melanoma (Table 5). Tumour stage recorded under the American Joint Committee on Cancer system was also strongly associated ($p = 8.74 \times 10^{-45}$, corrected $q = 5.15 \times 10^{-43}$). Stage therefore retains predictive value even inside a pooled analysis of four cancer types.

Among individual gene activity columns, DKFZp547D155 showed the strongest correlation with survival time ($\rho = -0.346$, corrected $q = 7.57 \times 10^{-31}$). Several genes recovered by reversing the compression also passed this screen independently. CA9 gave $\rho = -0.338$ (corrected $q = 7.23 \times 10^{-30}$) and CXCL5 gave $\rho = -0.308$ (corrected $q = 4.33 \times 10^{-25}$). The screen makes no use of the fitted model, so this agreement provides a check on the model-derived gene ranking.

Table 5: Outcome by cancer type across the pooled cohort ($N = 1,620$).

Cancer type	TCGA code	n	Death rate	Median survival (months)
Glioblastoma multiforme	GBM	595	0.827	12.16
Pancreatic adenocarcinoma	PAAD	184	0.543	15.34
Liver hepatocellular carcinoma	LIHC	372	0.355	19.76
Skin cutaneous melanoma	SKCM	469	0.475	35.78

5 Discussion

5.1 Why a linear model was chosen

A neural network could process all 500 candidate genes without a correlation filter or a compression step. On a cohort of this size that flexibility is expensive. With 972 training patients, a network large enough to capture interactions between genes has more trainable parameters than it has training examples. The usual defence is heavy regularisation, which in practice can push the network back towards a linear function.

The model reported here keeps a linear structure from the start. The correlation filter removed 441 redundant genes before they could destabilise the fitted weights. The smoothed elastic-net penalty then shrank the remaining correlated variables together, instead of selecting one representative from each group [8]. The resulting test concordance index of 0.736 is comparable to the 0.68 reported by an autoencoder model on 360 TCGA liver cancer patients [25]. The two numbers come from different cohorts and different splits, so this is a reference point rather than a controlled comparison.

Reversing the compression extends readability past the summary variables and back onto named genes. That computation is exact. Sampling-based explainers estimate the same quantity approximately [20, 21]. An exact decomposition is useful wherever the same explanation must be reproducible across repeated runs.

5.2 Clinical readability

A relative risk score can tell a clinician that one patient is at higher risk than another. It cannot answer what that patient’s survival probability is over a stated period. The calibration layer supplies that conversion, with Brier scores between 0.132 and 0.178 across the four horizons. The simulation layer then converts a probability into a range of plausible survival months.

The asymmetry of those ranges carries information. Bounds were tight for high-risk patients and wide for low-risk patients. This pattern suggests the model is confident about near-term decline in aggressive cases, and appropriately uncertain about long-horizon outcomes in more favourable cases.

Simulation was used rather than a confidence interval on the fitted weights. A confidence interval on β describes a population-level parameter. The simulation estimates a distribution for one patient’s survival time. Because the baseline hazard step function ends at the cohort’s longest observed follow-up of 218.4 months, the simulation cannot project beyond the available data.

5.3 Limitations

Six limitations should be stated.

The proportional hazards assumption was not tested. Schoenfeld residuals provide the standard check [47], and were not computed. The assumption is likely violated for at least some of the four cancer types, given their different progression timelines.

Both selected hyperparameters sat at the upper edge of their search grids. $\alpha = 3.0$ was the largest penalty strength searched, and $l1_ratio = 0.7$ was the largest balance value searched. A stronger penalty or a more lasso-weighted balance may perform better and was not explored.

Pooling four histologically different cancers into one weight vector averages gene effects across them. A gene that is protective in one cancer type and harmful in another will have those effects cancel. Subtype-specific precision is traded for coverage across cancer types.

The model has not been tested outside TCGA. Any use beyond this retrospective setting would need testing against batch effects and different sequencing platforms [23].

Each patient is represented by one tissue sample taken at diagnosis. Disease progression over time is therefore not modelled. Methods that use repeated measurements need data TCGA does not provide [24].

The statistical screen tested one variable at a time. It shows association with mortality and does not establish that any variable adds predictive value beyond the others.

6 Conclusion

A Cox proportional hazards model with an elastic-net penalty was fitted to clinical records and gene activity measurements from four pooled TCGA cancer cohorts. Redundant genes were removed first, reducing 500 candidates to 59, and those were compressed into 50 summary variables. The resulting 70-column model reached a concordance index of 0.736 on a held-out test split of 324 patients. Isotonic regression converted the raw risk scores into survival probabilities, with Brier scores as low as 0.132. A 5,000-draw simulation against the Breslow baseline hazard turned those probabilities into per-patient ranges of plausible survival months. Reversing the compression by matrix multiplication returned an exact risk weight for each of the 59 genes, so no approximate explainer was needed. A separate single-variable screen across all 1,049 columns recovered several of the same genes, including CA9 and CXCL5. Three steps are needed next. Interaction terms specific to each cancer type should be added. The pipeline should be tested on cohorts outside TCGA. The proportional hazards assumption should be checked with Schoenfeld residuals before any clinical use is considered.

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Disclosure of interest

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Declaration of generative AI use

[Insert the appropriate declaration of generative AI use here.]

Data availability statement

The data analysed in this study are publicly available from The Cancer Genome Atlas through cBioPortal [32, 33].

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